

Evidence-Gated, Contract-Centric Safe Adaptation for GKP Decoding under Drifting Noise

Evidence-synchronized note; snapshot through T6.19.3, 20 July 2026

Abstract

Approximate Gottesman–Kitaev–Preskill (GKP) quantum error correction must decode analog modular syndromes while the effective displacement, measurement, and control statistics drift. We reorganize an earlier CNN-assisted, FPGA-oriented dual loop around an execution contract rather than a learned winner. Static and adaptive MAP experts own logical-error-rate (LER) decisions; a causal regime posterior, event state machine, conservative fallback, and versioned A/B banks own tail-safety actions; and a fixed-point fast path owns deterministic source-to-action execution. CNN, teacher, and distilled-student modules are replaceable extensions behind the same admission boundary.

The central result is an evidence-gated outcome, including the stopping rules. In the V4 untouched smooth matrix, ROUTE-A reduced LER relative to the pilot-locked EWMA comparator by 2.1687×10^{-5} (paired 95% interval $[1.9003, 2.4548] \times 10^{-5}$), but lost to static joint MAP and Window MAP; only periodic drift survived family-wise testing. Abrupt/OOD tests established registered non-inferiority to EWMA, not a tail advantage, and static MAP was far better under calibration shift. The resulting full-paper gate was NO-GO. A subsequent causal-headroom audit then stopped V5 before implementation: the deployable held-out mixture gained only 0.459% over its nested baseline and the incremental action-space headroom was 0.02549%, far below the registered 12% entry gate. No V5 RTL or placement-and-routing result is therefore claimed.

Several bounded advantages remain valid in independent evidence lanes. The V4 integrated pre-board stack completed one million CXXRTL cycles with zero visible-field mismatch, while a source-bound static-MAP core reached 41.02–42.21 MHz in three post-route estimates with a six-cycle contract. In a separate 9.6-million-cycle multimode study, observed-only posterior-weighted CPD reduced aggregate LER relative to static Euclidean decoding by 0.064668 ($[0.064413, 0.064926]$), with all 32 seed clusters positive. A matched two-GKP-CNOT reproduction found analog-ML failure reductions of 31.16%, 58.07%, and 67.98% at 9, 12, and 13 dB. These algorithmic and pre-board results do not rescue the stopped V5 system claim: measured board latency, jitter, power, an exact same-task external FPGA comparator, and official GQF controller superiority all remain null or blocked. A final 206-cell auxiliary atlas passed 24/24 integrity gates without introducing a cross-lane score or changing the V5 and board verdicts.

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1 Scope, argument, and evidence cutoff

The one-sentence argument of this note is:

For repeated approximate-GKP correction, an evidence-gated dual loop makes LER, tail safety, deterministic execution, and learning extensions separately falsifiable: the main V4/V5 superiority route is currently stopped, while narrow V4, independent decoder, teacher–student, and pre-board engineering results remain usable only within their registered task and evidence layer.

The old task board and `docs/paper_notes/CNN_FPGA_GKP_theory_note_draft.tex` remain the historical source for the CNN+FPGA affine-calibration line. The current claim state is instead governed by `docs/new_task_board.md`. This note preserves useful old results as extension evidence without allowing them to override newer matched-comparison failures. The physical setting follows analog GKP syndrome decoding [1, 2]; the adaptive-decoder motivation is aligned with evidence that decoder calibration and prior mismatch affect logical performance [3, 4].

1.1 Terminology ledger

The following names are used consistently throughout the note.

Table 1: Evidence cutoff used in this note. “Done” means that the registered task produced a terminal artifact; it does not imply a positive verdict.

Surface	Status at cutoff	Consequence for this note
T6.5.1–T6.9.3 (V4)	Done except T6.9.2 Blocked	V4 passed restricted simulator/pre-board gates but ended <code>NO_GO_FULL_HIGH_LEVEL_PAPER</code> ; real-board fields remain null.
T6.10.1 / T6.15.5 (V5)	Done	Causal-headroom entry gate failed and V5 ended <code>NO_GO_V5_EARLY_HEADROOM_STOP</code> ; no V5 formal, RTL, or P&R evidence exists.
T6.16.1–T6.19.3 (Phase 6C)	Done	The 206-cell six-lane atlas passed 24/24 integrity gates; literature/reproduction/project-native lanes remain separate and cannot upgrade V4/V5.
T7.1.1	In Progress, excluded	The paper-level claim–evidence–boundary matrix was not terminal at this cutoff.

Table 2: Canonical terminology and authority boundaries.

Canonical term	Definition	What it does not mean
ROUTE-A safe adaptive dual loop	Full contract-centric system: MAP experts, causal routing, event/fallback logic, A/B banks, and fast path	A synonym for the CNN, teacher, student, or oracle
Static joint MAP	Observed-only strong software comparator using a fixed noise model	A universal GKP MAP decoder; the current LUT is source-bound
Window MAP / EWMA adaptive MAP	Observed-only adaptive MAP experts updated from completed causal windows	Hidden-state or per-scenario oracle tuning
Regime posterior	Causal HMM/event evidence used to route between validated expert banks	A direct logical correction or privileged truth channel
Tail-safety layer	Event FSM, health flags, fallback, leakage/reset handling, and integrity rollback	Tail-performance superiority; T6.7.2 establishes only locked-EWMA non-inferiority
Fast path	Quantized phase-LUT, event, frame, and bank action with a six-cycle contract	Measured end-to-end board latency
Learning extension	Legacy CNN, Feedback-GRAPE teacher, or distilled student behind the same contract	The required main-system winner
Hidden-state oracle	Nondeployable model reference evaluated in an isolated truth lane	A deployable comparator or a paper-level recovery optimum
V5 candidate family	A proposed continuation admitted only if causal deployable headroom clears the preregistered entry gate	An implemented system; V5 was stopped before formal/RTL work
Auxiliary evidence lane	A task-local reproduction or project-native matched result with an explicit evidence grade	Evidence that can rescue or reopen the V4/V5 promotion gate
Pre-board estimate	Fixed-point/CXXRTL/synthesis/P&R result without a physical-board measurement	Measured latency, jitter, power, transport, or HIL

2 Why the architecture changed without discarding the old work

The earlier CNN-centric line produced a useful design pattern: a slow estimator updates a low-dimensional correction surface, while the latency-critical path uses a small quantized rule. It also produced a frozen four-scenario result in which a teacher-anchored residual branch outperformed UKF. However, later evidence exposed three limitations.

First, the learned strong branch did not survive the unified evidence gate. T5.1.4 found no matched learned decoder in the registered matrix, no Holm discoveries across 24 family tests, and a calibration-shift transient in which the adaptive candidate had 55 errors per 512 decisions against 37 for the reference. Second, a generic uncertainty fallback had a positive aggregate OOD effect but significantly harmed the compound family and imposed a small nominal cost. Third, the first unified ROUTE-A integration run was much worse than Window/EWMA because the policy froze or reverted the wrong state under tail evidence.

These are not reasons to discard the affine fast path, the historical learned results, or the hardware work. They are reasons to change the unit of contribution. The new unit is an auditable contract that assigns each claim to the module and evidence lane that can actually support it.

3 Positioning relative to GKP decoding and feedback control

Analog GKP decoding exploits more information than hard binning because the wrapped residual carries distance-to-boundary and noise-model information [1, 2]. This motivates both joint MAP rules for correlated quadratures and analog likelihoods passed to an outer-code decoder. Our static, adaptive, and CPD lanes share that statistical premise, but they answer different tasks: single-mode repeated correction, gate-level CNOT failure, structured outer-code decoding, and multimode lattice decoding are never collapsed into one leaderboard.

Calibration-aware QEC emphasizes that a decoder can become mismatched when priors or device statistics move [3, 4]. The present work extends this observation into a systems contract: adaptation is causal, versioned, bounded, and tested against static and multiple adaptive baselines; tail non-inferiority and update actions are reported separately from mean LER. This differs from treating an adaptive estimator as automatically beneficial whenever a truth oracle exposes nonzero headroom.

Non-Markovian feedback and experimental real-time QEC demonstrate that memory and control policies can improve finite-horizon performance in suitable physical protocols [5, 6]. Here those results motivate a teacher-student extension and an AQEC comparison lane, but literature facts are not promoted to project reproduction. The contribution is therefore the evidence-bounded integration and falsification framework, together with the task-local decoder and pre-board results that survive it.

4 Contract-centric dual-loop method

4.1 System overview and reuse of the dual-loop pipeline

Figure 1 is retained from the presentation material because its interface structure survives the move away from a CNN-centric claim. The durable elements are observed syndrome history, a bounded slow-loop proposal, safe publication of a complete parameter image, versioned A/B banks, a quantized MAP-LUT, an event state machine, and a Pauli frame update. These interfaces, rather than the identity of one learned estimator, define the reusable architecture.

The fast loop consumes the current validated bank on every measurement and never invokes the offline teacher. The slow loop may host analytical, filtering, or learned estimators, but each candidate must emit the same bounded image and pass the same admission, integrity, and rollback rules. The diagram therefore acts as an interface map, not as evidence that the four-state student is the main LER winner or that a complete FPGA deployment has been measured.

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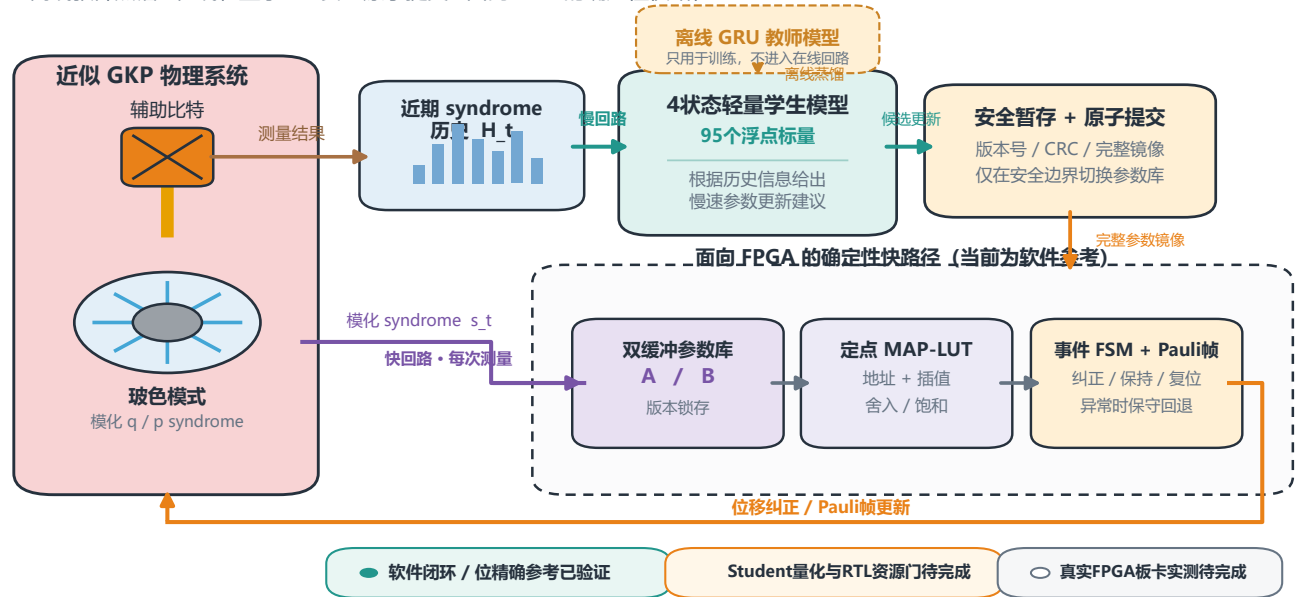


Figure 1: Reused Chinese presentation schematic of the drift-adaptive GKP dual loop. Under the current ROUTE-A interpretation, the upper four-state student is an optional learning extension; Window and EWMA adaptive MAP provide the primary deployable parameter images, while HMM/event/fallback logic controls safe routing and publication. The maturity badges in this 16 July 2026 presentation asset are an archival snapshot: subsequent tasks added student fixed-point RTL and pre-board resource evidence, but real-board validation remains pending.

4.2 Task formulation

At cycle t , the deployable decoder receives only quantized observed syndrome packets and health/event fields. It returns a logical action, frame update, fallback reason, active-bank identity, and versioned provenance. Truth fields used to generate trajectories are denied to all deployable adapters and are available only to the isolated evaluator and model oracle.

For each completed parameter window, the host side maintains two causal expert estimates:

$$\hat{\theta}_t^W = \text{Window}(y_{1:t}), \quad \hat{\theta}_t^E = \text{EWMA}(y_{1:t}), \quad (1)$$

where $y_{1:t}$ denotes observed syndrome history only. Each estimate compiles to a parameter-bank image for the same fixed-point fast path. The router may promote the Window bank only when the causal smooth posterior, pre-update predictive score, and hysteresis checks all pass. Tail or uncertain evidence uses the continuously updated and validated EWMA shadow; integrity faults alone may republish the last-known-good image.

4.3 Unified execution contract

Table 3: Frozen execution fields shared by deployable methods.

Field	Frozen value	Interpretation
Syndrome quantization	10-bit ADC; 8-bit LUT address; 2-bit fraction	Same quantized packet domain for every deployable comparator
MAP-LUT word	Signed Q9.12, ties-to-even, saturation	Deterministic arithmetic and packed-word behavior
Fast path	5-cycle MAP plus 1-cycle event/action	Six-cycle logical action-valid contract
Bank protocol	CRC/SHA/CAS A/B bank, versioned commit, six-cycle drain	No silent partial publication; integrity-only LKG rollback
Cadence	32-cycle regime update; 2,048 scalar samples; 4,000-cycle parameter update	Completed-window causality and explicit update age
Matched ceilings	8,192 update MACs; 8,192 B state; 8,192 B workspace; 5,000 μ s host update	Fail-fast budget admission, with actual use still reported
Oracle isolation	Separate exact-key truth packet	No deployable aggregate may consume hidden state

The current hardware LUT is phase-conditioned and is not mathematically equivalent to the full two-dimensional joint MAP used as a strong software comparator. The latter is therefore retained in the comparison table but is not relabelled as current-RTL deployable.

4.4 Safety state and atomic updates

The V4 policy replaced two failed architectures. A static-switch design and a freeze-all-EWMA design both failed all 38 pilot-safe candidate tuples because the supposedly trusted state became stale under persistent calibration or telegraph changes. V4 instead updates Window and EWMA shadow banks continuously. A 20,061-cycle structural trace produced versions 1–4 at cycles 4001, 8002, 12003, and 16005; all source-to-action fields were six cycles. At the tail transition, the EWMA shadow was committed rather than reverting to static. An integrity fault cancelled a pending Window update and completed an LKG rollback. The registered dual-shadow update cost was 1,218 MACs under the 8,192-MAC ceiling.

This trace proves structural execution and provenance, not tail-performance superiority. The independent T6.7.2 matrix subsequently established bounded non-inferiority to locked EWMA, as reported below.

4.5 Replaceable learned modules

Learning is outside the critical claim path. A legacy CNN can propose a bounded update, a non-Markovian teacher can supply an offline control target, and a low-dimensional student can compress that target. All three must obey the same schema, cadence, bank, and failure rules. In the unified runner, the legacy CNN required 3,489,984 MACs, 1,586,368 B of state, and at least 65,536 B of workspace, so it was correctly demoted to ablation-only under the frozen budget. This failure does not invalidate the contract; it demonstrates why learned modules must be replaceable rather than definitional.

5 Evaluation protocol and metrics

5.1 Result-blind formal design

The formal protocol was frozen after disclosure of earlier counterexamples but before access to the new formal split. Calibration, pilot, and formal data use 12, 12, and 24 independent seed clusters. The complete protocol contains 143 cells and 71,958,528 formal decisions per method across smooth and abrupt/OOD families. T6.7.1 consumed the smooth subset: four families, six held-out cells per family, 24 formal clusters, and 576 trajectories. T6.7.2 then consumed the locked abrupt/OOD and nominal subset: six fault families, one nominal cell, 24 formal clusters, and 888 trajectories. Together the two tasks exhaust the pre-registered formal decision count without retuning the V4 lock.

All methods receive the same quantized trace. The primary comparator was selected once on pilot data and frozen as EWMA. Formal results cannot trigger baseline reselection, per-family threshold changes, family deletion, or router retuning. Inference uses 20,000 paired bootstrap replicates with the formal seed as the independent cluster and equal family weighting. Family-specific discoveries use Holm correction.

5.2 Metrics

The Pauli-resolved logical error rate is

$$p_L = p_X + p_Y + p_Z = 1 - p_I. \quad (2)$$

The registered primary smooth contrast is

$$\Delta_{\text{EWMA}} = p_L(\text{EWMA}) - p_L(\text{ROUTE-A}), \quad (3)$$

so a positive value favours ROUTE-A. The static-to-oracle gap closure is

$$G_{\text{static}} = \frac{p_L(\text{static}) - p_L(\text{ROUTE-A})}{p_L(\text{static}) - p_L(\text{oracle})}. \quad (4)$$

Negative G_{static} means that ROUTE-A is worse than static MAP; it is not clipped to zero. Tail readouts use non-overlapping 512-decision windows, reporting p95 and global worst-window LER. Action diagnostics report fallback, unnecessary fallback, avoided and induced errors, false updates, commits, and adaptation lag separately from average LER.

5.3 Evidence grades and stopping rules

Results are assigned to one of five non-interchangeable layers: literature fact, official-code reproduction, project-native matched simulation, fixed-point/CXXRTL/P&R pre-board evidence, or measured board/HIL evidence. Missing task adapters, source artifacts, checkpoints, or timing boundaries are reported as N/A, null, blocked, or ineligible; they are not imputed from a neighbouring lane. In particular, a positive Phase 6C decoder result cannot upgrade the V5 system gate, and a P&R estimate cannot populate a board-measured field.

V5 also used a causal entry gate before expensive implementation. Let $H_{mnested}$ be the improvement of a deployable selector over the best nested baseline on development data and $H_{maction}$ the additional improvement made possible by expanding the correction-action family rather than by truth-privileged switching. The preregistered continuation conditions included positive nested evidence and

$$H_{maction} \geq 12\%. \quad (5)$$

Failure stops formal evaluation, quantization, RTL, and P&R for that candidate family. This protects downstream engineering from turning a diagnostic oracle gap into a deployable claim.

6 Results

6.1 Legacy CNN+FPGA evidence retained as an extension lane

The earlier T24 frozen-set benchmark remains valid within its original mock-backed software-HIL scope. Hybrid Residual-B was the winner and UKF the runner-up in all four scenarios.

Table 4: Historical T24 result retained as learning-extension evidence. The relative reduction is $(p_L^{\text{UKF}} - p_L^{\text{hybrid}})/p_L^{\text{UKF}}$. This table is not part of the new unified ROUTE-A ranking.

Scenario	UKF final LER	Hybrid final LER	Relative reduction
Static bias+ θ	0.825370	0.810902	1.75%
Linear ramp	0.811201	0.787755	2.89%
Step $\sigma+\theta$	0.811548	0.788800	2.80%
Periodic drift	0.821558	0.806392	1.85%

Two later scenario-specific repeat sets strengthened only the direction of this historical result. With 12 paired formal-length repeats, the mean UKF-minus- hybrid differences were 0.015563 for static bias (paired- t 95% interval [0.014778, 0.016348]) and 0.022417 for linear ramp ([0.021215, 0.023619]); all 12 pairs were positive in both scenarios. The other two scenarios, all-scenario pooled inference, held-out robustness, and hardware validation were not closed by that evidence.

The proper reuse is therefore modest: a learned residual can be useful on a bounded affine parameter surface. It is not evidence that CNN is the best slow-loop estimator, that it satisfies the new matched budget, or that the complete system wins in tail and latency.

6.2 Untouched smooth formal comparison

Table 5 reports the complete T6.7.1 method ordering. It contains the result that controls the current LER claim.

The primary contrast passed:

$$\Delta_{\text{EWMA}} = 2.1687 \times 10^{-5}, \quad \text{CI}_{95\%} = [1.9003, 2.4548] \times 10^{-5}. \quad (6)$$

This is a 2.14% relative reduction from the locked EWMA LER. The gain was not uniform across smooth families.

Against stronger descriptive comparators, the conclusion reverses. ROUTE-A is worse than static joint MAP by 2.45×10^{-5} and worse than Window MAP by 9.63×10^{-5} . The static-to-oracle gap closure is

$$G_{\text{static}} = -0.03046, \quad \text{CI}_{95\%} = [-0.04966, -0.01119]. \quad (7)$$

Thus the current smooth result supports a pre-registered EWMA contrast and a periodic-drift effect. It does not support “best deployable decoder”, “better than static GKP decoding”, or “general smooth-drift advantage”.

Table 5: T6.7.1 smooth formal result. Each method has 28,311,552 scored decisions. The oracle is nondeployable. Window statistics use 512 decisions.

Method	Aggregate p_L	p95 window LER	Worst window LER
Standard binning	9.7861×10^{-4}	3/512	17/512
Static joint MAP	9.6819×10^{-4}	3/512	16/512
Window MAP	8.9642×10^{-4}	2/512	64/512
EWMA adaptive MAP	1.01443×10^{-3}	2/512	59/512
Kalman adaptive MAP	1.78574×10^{-3}	3/512	85/512
ROUTE-A	9.92740×10^{-4}	2/512	59/512
Hidden-state oracle	1.62337×10^{-4}	1/512	7/512

Table 6: Family-specific EWMA-minus-ROUTE-A contrasts.

Family	Difference	Paired 95% interval	Holm discovery
Mean drift	5.65×10^{-7}	$[-2.83 \times 10^{-7}, 1.41 \times 10^{-6}]$	No
Variance drift	0	$[-7.06 \times 10^{-7}, 7.06 \times 10^{-7}]$	No
Correlation drift	0	$[0, 0]$	No
Periodic drift	8.618×10^{-5}	$[7.545, 9.735] \times 10^{-5}$	Yes

6.3 Action and update diagnostics

The formal action ledger contained 884,736 scored posterior updates and 13,824 commits. Of those commits, 2,870 used the Window bank and 10,954 used the EWMA bank. There were 661 avoided errors and 47 induced errors relative to EWMA, with zero false updates under the smooth definition. However, the fallback signal rate was 21.6305%, and the unnecessary-fallback decision rate was 21.5822%. Every trajectory had an adaptation lag of 1,904 decisions.

The net avoided-minus-induced count is encouraging for the locked smooth contrast, but the high fallback rate shows why average LER cannot substitute for a tail-safety experiment. Earlier T5.4.2 evidence reinforced this caution: generic uncertainty gating reduced aggregate OOD error by 0.00107490 with a 95% interval $[0.00001950, 0.00227615]$, yet it significantly harmed the compound family by -0.00488536 and imposed a small nominal cost. T6.7.2 therefore reported every abrupt/OOD family and the nominal negative control without using the periodic smooth gain to offset a failure.

6.4 Abrupt/OOD and nominal non-inferiority

The independent T6.7.2 matrix contains 888 trajectories and 43,646,976 scored decisions per method across step, telegraph, burst, readout/reset, leakage, compound, and nominal conditions. All registered catastrophic-degradation, calibration-strict, and nominal non-inferiority gates passed. The result must nevertheless be read as a safety bound against the locked EWMA baseline rather than as a decoder-performance gain: step, telegraph, readout/reset, leakage, and compound had exactly zero ROUTE-A–EWMA difference in all core safety statistics. Burst was aggregate-neutral, with a maximum single-window excess of one error and two avoided versus two induced errors.

The calibration result removes the earlier degradation only relative to locked EWMA: both methods have a worst window of 181/512 and a seed-worst paired difference of 0 $[0, 0]$. Static MAP is substantially better at 32/512 and lower average LER. Nominal ROUTE-A–EWMA average difference and induced-minus-avoided count are also 0 $[0, 0]$.

The operational cost is large in the tail cells. Fallback signals span 59.45–95.85%, unnecessary fallback spans 58.80–94.63%, and 2,044–3,365 of 3,456 commits per family occur inside the pre-registered tail truth interval and are therefore counted as false updates. V4 blocks Window promotion under tail evidence

Table 7: T6.7.2 calibration-shift and nominal results. Equality to EWMA passes the registered non-inferiority gates but is not an advantage over that comparator. Window statistics contain 512 decisions.

Method or diagnostic	Average LER / rate	Global worst / bound
Static joint MAP, calibration shift	0.01087	32/512
Window MAP, calibration shift	0.01243	182/512
Locked EWMA, calibration shift	0.01227	181/512
ROUTE-A, calibration shift	0.01227	181/512
Nominal fallback	0.11936%	registered bound 1.0%
Nominal unnecessary fallback	0.11783%	registered bound 0.75%

but continues to update and publish the trusted EWMA shadow; it must not be described as freeze-all adaptation. Detection had no censored events (burst p95 96 decisions), and recovery-to-OPEN also had no censoring in the recoverable families (p95 256–320 decisions). These diagnostics support auditable behavior and registered non-inferiority, while directly ruling out claims of lower tail LER, low fallback, or superiority to static MAP.

6.5 Deterministic execution and pre-board hardware evidence

The latency advantage currently established is determinism and feasibility, not measured superiority over another decoder.

Table 8: Hardware-facing evidence layers. P&R numbers are open-tool estimates, not vendor signoff or board measurements.

Surface	Quantitative result	Supported interpretation
Source-bound static-MAP core	Three-seed Fmax 41.024–42.212 MHz; at most 3,387 LUT4, 865 FF, 8 BRAM and 2 DSP; six cycles and initiation interval one	A compact current-source core clears the 27 MHz contract clock in post-route estimate
Integrated V4, no student	Minimum Fmax 39.137 MHz; at most 3,859 LUT4, 1,069 DFF and 8 BSRAM; six-cycle source-to-action path	The contract, router, bank and action path fit together in the selected open-tool target
Integrated V4 plus 4-state student	Minimum Fmax 38.506 MHz; at most 4,889 LUT4, 1,210 DFF and 8 BSRAM	A compressed replaceable student remains feasible outside the per-shot critical path
Integrated CXXRTL qualification	4 smooth plus 6 fault families, each 100,000 cycles; zero visible-field mismatch, undefined action, route/core CRC error, or silent overflow; 19/19 gates	Board-independent bit-accurate execution and recovery qualification

The integrated qualification also exercised six modes, five health states, three actions, nine reachable fault bits, CRC/version failures, FIFO disturbance, commit rejection, and counter/LLR saturation. The HMM itself remains a software slow-loop component; the synthesized design consumes the quantized posterior/policy overlay. Formal replay was 99.5802% real traces and 0.4198% directed branch vectors. This evidence does not cover real transport, clock-domain crossing, metastability, bitstream generation, power, ADC/DAC latency, or board deadline distributions. No measured FPGA latency, end-to-end latency, or external fastest-decoder claim is active.

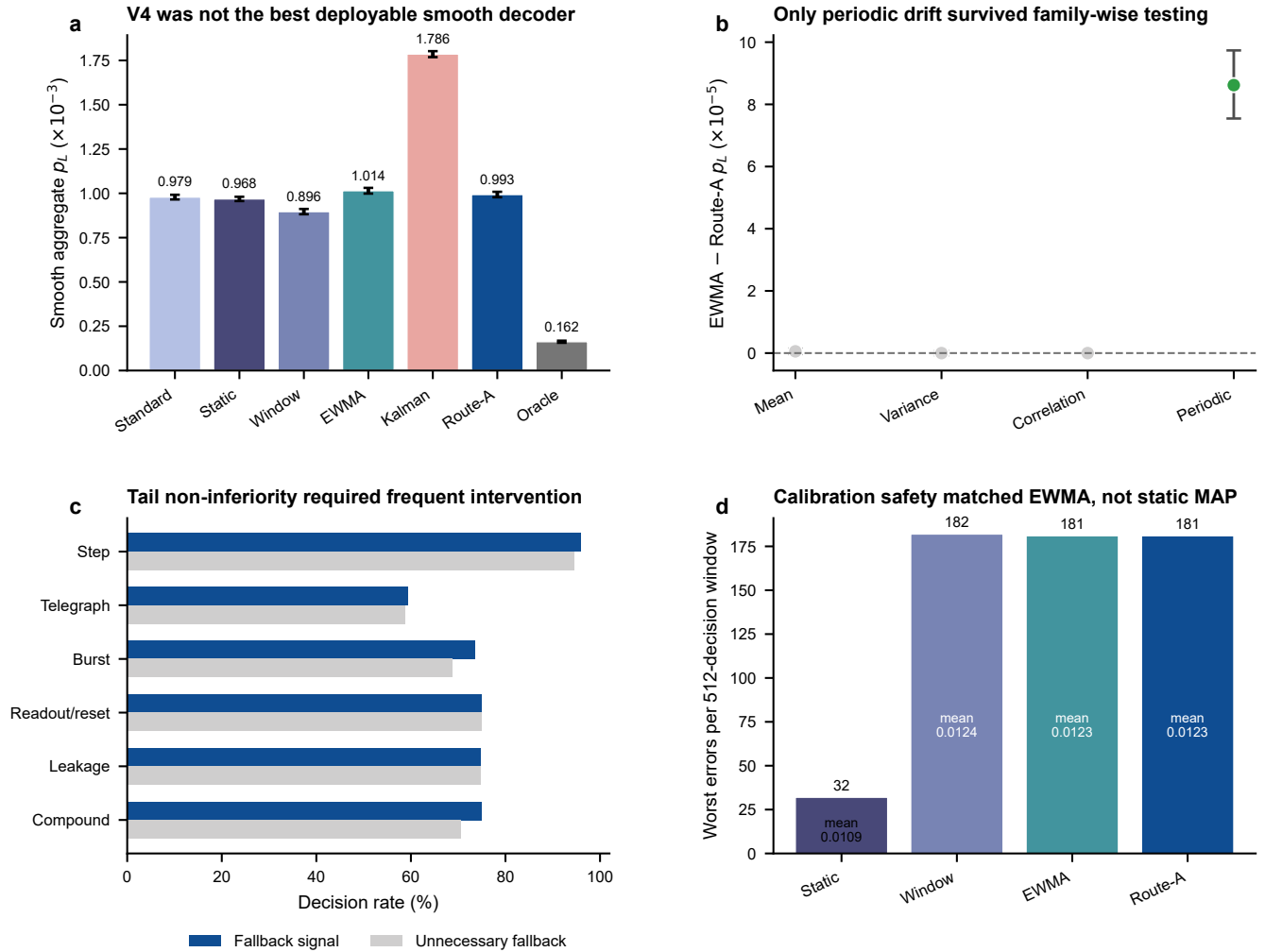


Figure 2: The V4 result is bounded rather than globally positive. (a) Equal- family smooth LER places Window and static MAP below ROUTE-A; the oracle is truth-privileged. Error bars are seed-cluster 95% intervals. (b) Only the periodic EWMA-minus-ROUTE-A contrast survived Holm correction. (c) Tail non-inferiority required frequent fallback and unnecessary-fallback signals. (d) Under calibration shift, ROUTE-A matched EWMA at 181 errors in the worst 512-decision window, while static MAP had 32. Source: frozen T6.7.1 and T6.7.2 machine artifacts; plotted-source rows and hashes accompany the figure.

6.6 Causal-headroom audit and V5 early stop

The V4 NO-GO triggered a causal audit before any V5 model or RTL was built. Across 71,958,528 diagnostic expert decisions, the best deployable aggregate baseline was EWMA with $p_L = 0.01154953$. Truth-privileged selectors appeared to offer substantial improvement—2.4737% at family level, 4.1784% at cell level, 16.665% when activation truth was exposed, and 29.523% per decision—but these values use information unavailable to an online decoder.

The deployable audit used 186 development trajectories and 4,571,136 decisions. A strict causal selector produced 35,396 errors against 35,314 for its nested baseline, or -0.2322% headroom. A separately frozen held-out mixture produced 35,152 errors, only 0.4587% better than the nested baseline. More decisively, the hard decision oracle had 29,105 errors and the expanded action oracle had 29,096: nine errors separated them, corresponding to only 0.02549% incremental action-space headroom versus the 12% entry gate.

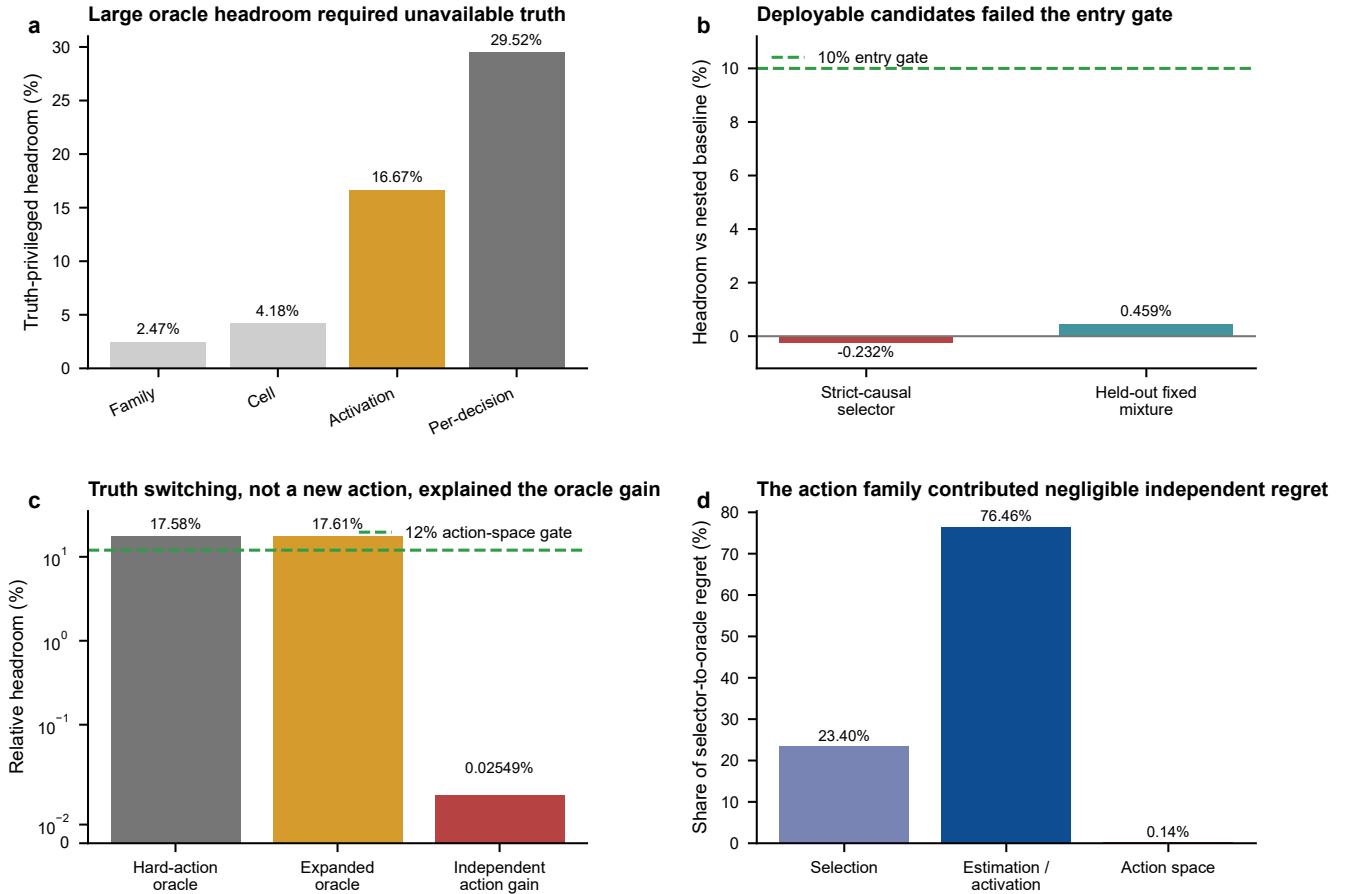


Figure 3: Why V5 was stopped before implementation. (a) Large diagnostic oracle headroom grows with access to unavailable truth. (b) Deployable selectors do not approach the registered 10% entry level. (c) Hard and expanded truth oracles are almost identical; the new action family contributes 0.02549%, far below the 12% action-space gate. The logarithmic axis retains the small positive value. (d) The selector-to-oracle regret is dominated by estimation/activation (76.46%), not action-space capacity (0.14%). Source: T6.10.1 causal-headroom JSON/CSV; percentages are independently recomputed by the plotting script.

The resulting verdict was `NO_GO_V5_INSUFFICIENT_ACTION_SPACE_HEADROOM`, later frozen as `NO_GO_V5_EARLY_HEADROOM_STOP`. Tasks T6.10.2–T6.15.4 were dropped. Consequently there is no V5 untouched LER matrix, tail test, quantized-retention result, formal proof, CXXRTL run, or P&R profile. Reusing V4 hardware artifacts as if they belonged to V5 would violate the evidence contract. The early stop is therefore a substantive negative result: the observed oracle gap primarily calls for better causal state estimation or a

different physical task, not a larger correction-action menu.

6.7 Teacher–student extension evidence

The non-Markovian teacher/student line remains useful as a replaceable control extension motivated by feedback-based QEC and real-time experimental control [5, 6]. Figure 4 reuses the presentation result panel because it exposes both comparator ordering and the paired-seed sample structure. The metric is a ten-cycle, finite-horizon area-equivalent effective fidelity lifetime; it is neither an asymptotic decay constant nor FPGA wall-clock time.

轻量学生模型基本保留教师模型的有效保真寿命

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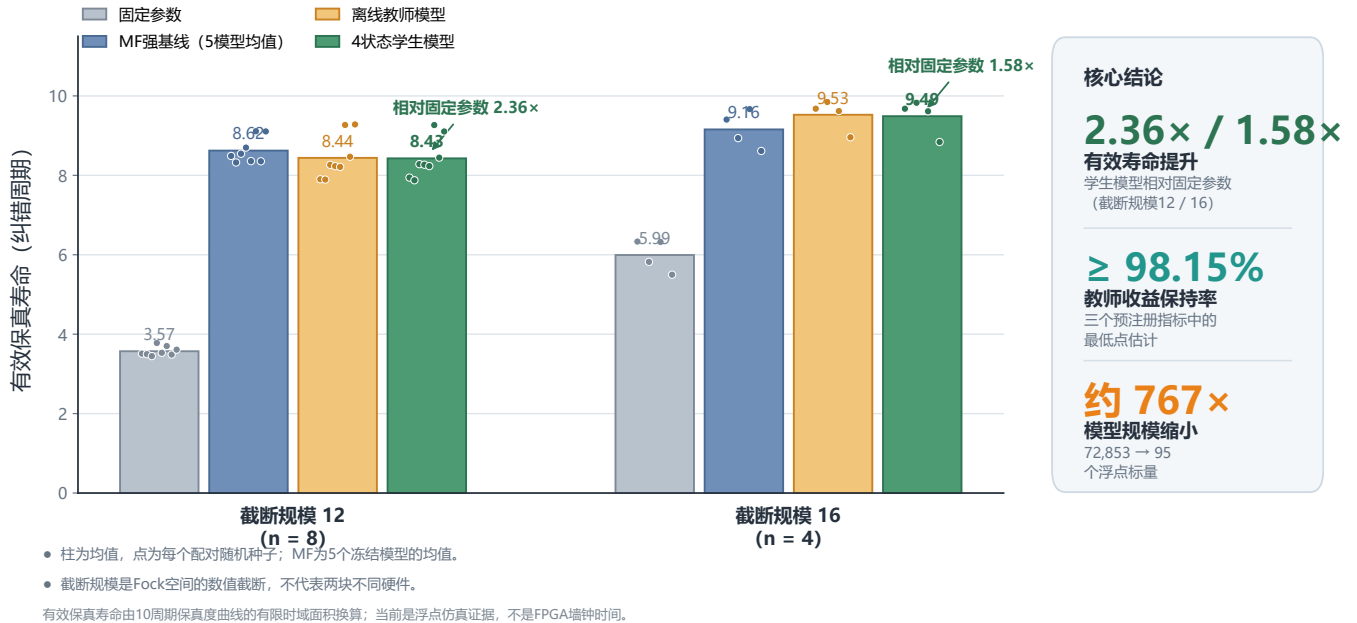


Figure 4: Finite-model teacher–student extension evidence. At cutoff 12 ($n = 8$ paired evaluation seeds), the four-state student reached an effective fidelity lifetime of 8.427 cycles versus 3.571 for fixed parameters, a $2.36\times$ ratio. At cutoff 16 ($n = 4$), the corresponding values were 9.489 and 5.994 cycles, a $1.58\times$ ratio. Bars show seed means and points show paired seeds; the MF bar is the mean of five frozen agents, not a test-selected winner. The sidebar’s 98.15% is the lowest point retention, whereas the lowest paired-bootstrap 95% interval lower bound is 94.45%. All quantities are float finite-model results.

The student closely tracked the offline teacher: their cutoff-12 lifetimes were 8.427 and 8.440 cycles, and their cutoff-16 lifetimes were 9.489 and 9.525 cycles. The five-agent MF mean was slightly higher at cutoff 12 (8.623 cycles) and lower at cutoff 16 (9.156 cycles), so the figure does not support universal teacher superiority. Across the three preregistered retention metrics, the lowest student point retention was 0.981457 and the lowest 95% interval lower bound was 0.944501 relative to teacher-versus-standard gain.

The selected student uses four recurrent states and 95 stored scalars, versus 72,853 for the teacher, a $\sim 767\times$ float-model compression. Its fixed-point RTL produced zero mismatches across 7,680 outputs, with maximum fixed-versus-float error 1.46038×10^{-4} .

This is evidence for gain retention and compression within the project model. It is not an official reproduction or improvement over Puviani *et al.*, not a general memory-mechanism result, and not the main decoder LER claim. The larger quantized GRU lower-bound required 72,854 cycles (2,698.30 μs at 27 MHz), exceeded the five-microsecond slot, and was dropped. That negative result supports the architectural decision to keep learning outside the critical fast path.

6.8 Independent Phase 6C decoder and comparison lanes

After the V5 stop, Phase 6C was executed as a read-only auxiliary programme. Its split and hashes cannot enter the V5 denominator, and each result keeps its own task and evidence grade. This separation matters because the strongest new algorithmic result is positive even though the system-promotion result is not. The final atlas contains 206 cells and passed 24/24 integrity gates, 24/24 single-gate mutations, and 162 joint regressions. Passing means that provenance, metric recomputation, null/negative states, and evidence boundaries are intact; it does not mean that every lane has a positive result.

Table 9: Phase 6C evidence lanes through T6.19.2. Rows are intentionally not combined into a global score.

Lane	Evidence grade	Result	Allowed interpretation
Two-GKP CNOT CI/ML	Project-native matched reproduction	3,080,192 trials; ML reduced gate failure by 31.160%, 58.065%, and 67.982% at 9, 12, and 13 dB; maximum literature discrepancy 1.52%	Analog ML helps for the reproduced matched gate model
Structured surface-GKP CPD	Official-code reproduction	2,005/2,005 upstream checks; small-distance crossing 0.599298 versus 0.600245; CPD lower LER in 27/27 reproduced cells	Partial crossing/ordering reproduction, not the full reported distance range
Multimode posterior-weighted CPD	Project-native matched	9.6 million cycles, 38.4 million modal decodes; adaptive $p_L = 0.172261$ versus static Euclidean 0.236929	Observed-only weighting gives a large drift gain in this independent model
AQEC lifetime replay	Project-native negative / official blocked	Active QEC was worse than idle in all six project-native wall-clock cells; official experimental protocol unavailable	Negative project model, not a refutation of the experiment
Learned-model eligibility	Ineligible	0 of 16 candidate families met the same-task contract; one legacy CNN replayed only as an exact diagnostic	No learned same-task ranking
GQF controller	Blocked	Exact source completeness 0/15; all 13 matched comparison metrics null	No reproduction or superiority over Puviani <i>et al.</i>
External FPGA decoders	Literature-only normalization	18 implementations normalized; exact same-task comparator count 0	No raw-ns ranking, fastest, or SOTA claim

For the multimode task, static posterior weighting alone was harmful in the aggregate ($p_L = 0.261679$), whereas the observed-only adaptive rule nearly matched the metric oracle (0.172261 versus 0.171929). Its absolute improvement over static Euclidean decoding was

$$0.0646684, qquad CI_{95\%} = [0.0644128, 0.0649264], \quad (8)$$

and all 32 seed clusters favoured the adaptive lane. The worst-window LER fell from 0.320313 to 0.291016 and CVaR₉₅ from 0.277271 to 0.240627. These are the clearest current LER and tail-metric advantages, but they belong to the independent multimode CPD task rather than to the stopped V5 policy.

7 Where the current data show an advantage

Table 10 is the compact answer to which metrics currently show an advantage. Rows are deliberately not collapsed into a global score because decoder, controller, and hardware lanes have different objects and units.

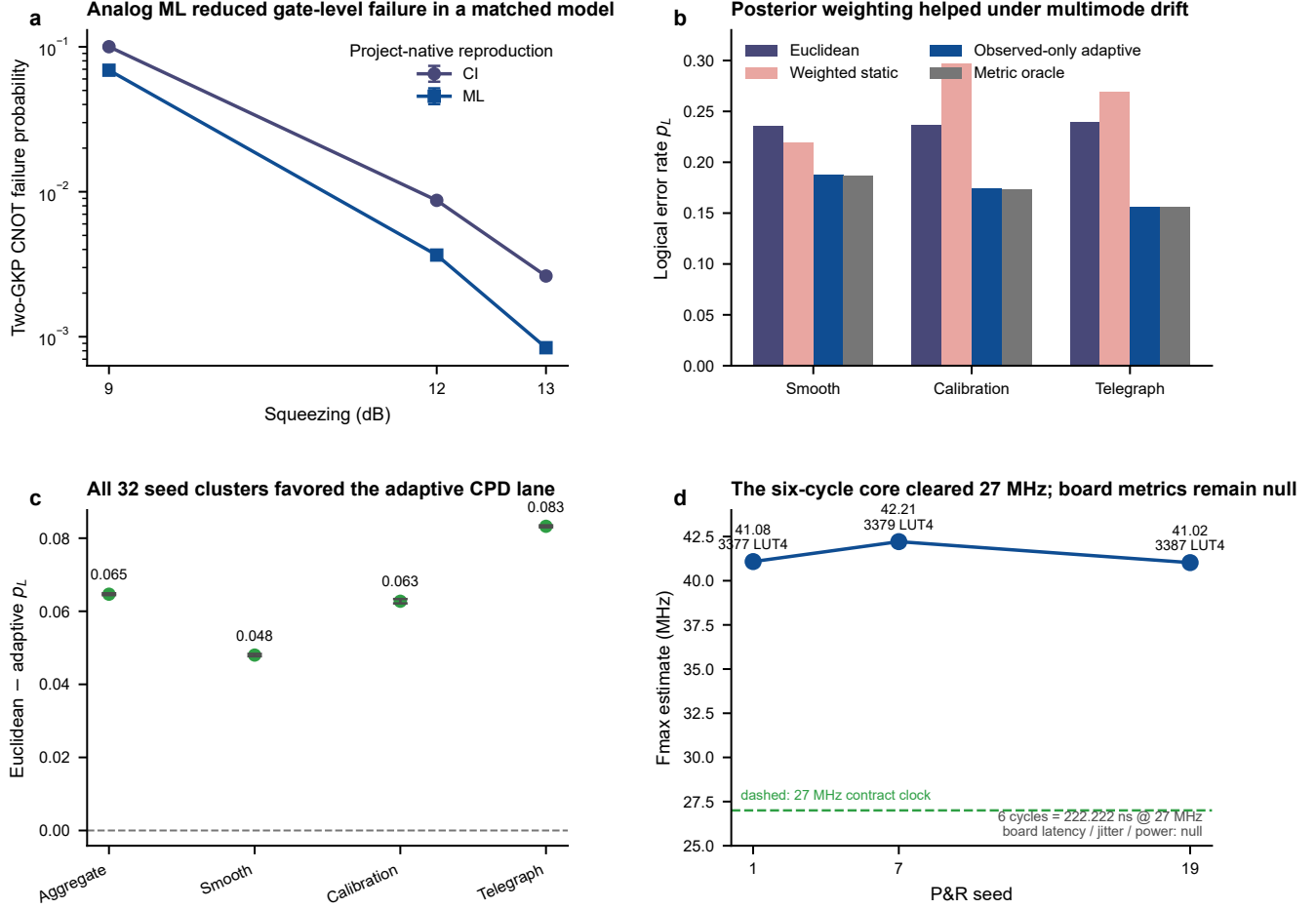


Figure 5: Bounded positive evidence outside the stopped V5 route. (a) Matched two-GKP-CNOT CI/ML reproduction; points show binomial estimates and 95% intervals. (b) Scenario LERs for the independent multimode CPD task. (c) Static-Euclidean-minus-adaptive LER is positive in the aggregate and each family; error bars are paired seed-cluster 95% intervals. (d) A source-bound six-cycle static-MAP core clears 27 MHz in three P&R seeds, but 222.222 ns is a contract-clock estimate and board latency, jitter, and power remain null. Sources: T6.17.2, T6.18.3, and T6.19.1 machine artifacts.

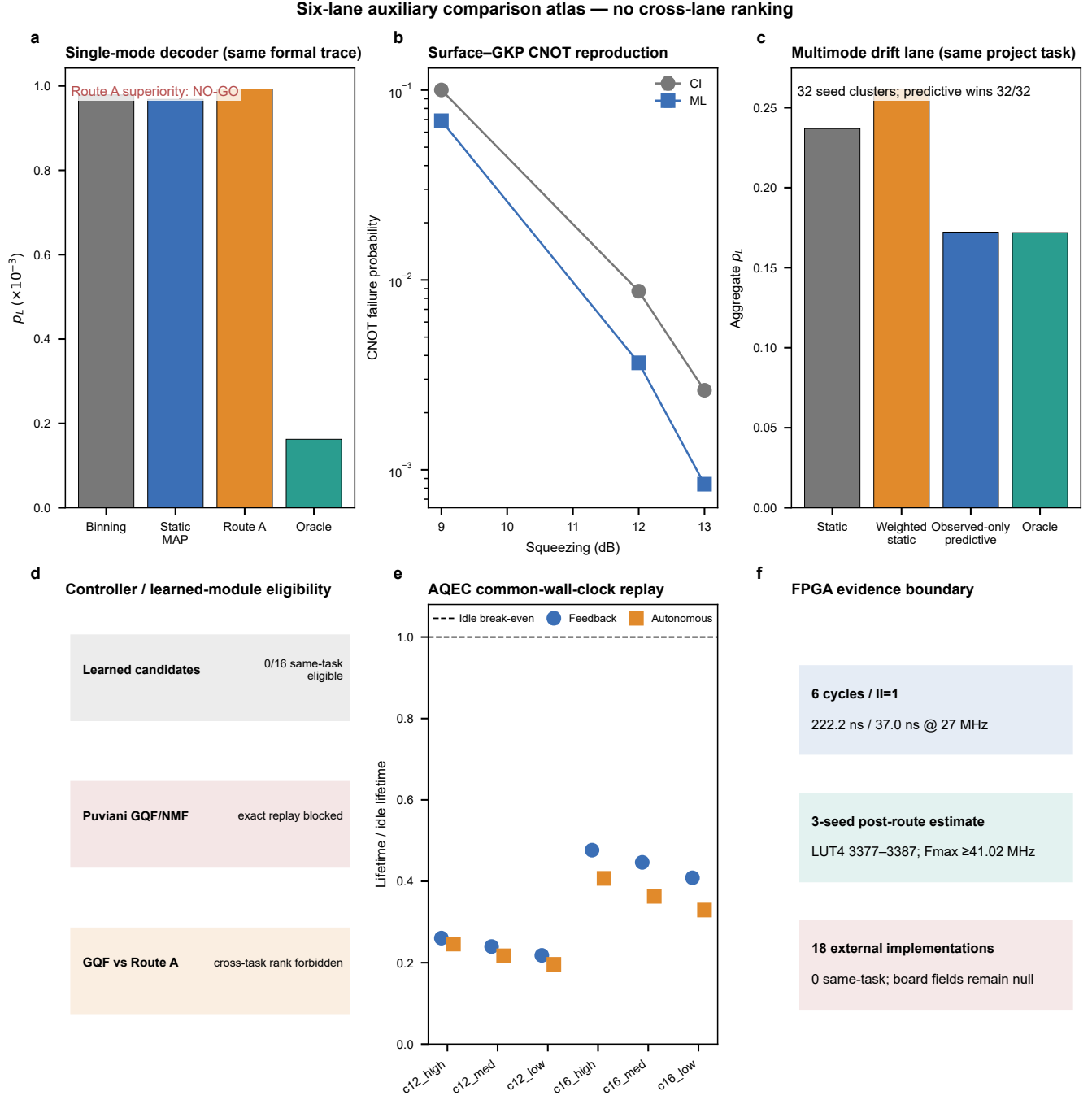


Figure 6: Integrity-gated six-lane auxiliary atlas. The panels retain, rather than hide, a single-mode Route-A NO-GO, a positive CNOT ML reproduction, a positive multimode observed-only CPD result, zero same-task learned/controller eligibility, six negative AQEC wall-clock cells, and FPGA estimate/null boundaries. The 206 cells bind report, raw source data, frozen configuration and SHA-256. The atlas has no global winner and cannot upgrade the V5 early stop or the blocked real-board gate.

Table 10: Metric-level advantage matrix at the T6.19.3 evidence cutoff.

Metric / property	Status	Evidence-supported statement	Boundary
V4 smooth LER vs locked EWMA	conditional	ROUTE-A reduced aggregate LER by 2.1687×10^{-5} , 2.14% relative, with a positive paired interval	Only the preregistered EWMA contrast; only periodic drift survived Holm correction
V4 LER vs static / Window	not established	ROUTE-A LER is higher than static and Window; static-to-oracle gap closure is negative	Direct counterevidence to a general V4 decoder advantage
V4 abrupt/OOD vs locked EWMA	conditional	All catastrophic, calibration-strict, and nominal non-inferiority gates passed	Five families are ties; burst is aggregate-neutral; this is not a tail advantage
V4 tail LER vs static MAP	not established	Calibration worst was 181/512 for ROUTE-A versus 32/512 for static MAP	High fallback and false-update rates remain visible
V5 deployable headroom	not established	Nested causal selector -0.2322% ; held-out mixture $+0.4587\%$; new action family only 0.02549% versus 12% gate	V5 stopped before formal, quantized, RTL, or P&R work
Multimode CPD LER and tail	supported	Observed-only adaptive CPD reduced aggregate LER by 0.0646684 with positive 95% interval; all 32 clusters positive; worst-window and CVaR ₉₅ improved	Independent project-native multimode task; cannot upgrade V5
Two-GKP CNOT analog ML	supported	Failure reductions of 31.160%, 58.065%, 67.982% at 9, 12, 13 dB across 3,080,192 trials	Matched gate-level reproduction, not a memory or hardware result
Structured surface-GKP CPD	conditional	Official-code small-distance crossing reproduced; CPD lower LER in 27/27 cells	Partial distance range; no full threshold or project hardware claim
Fast-path determinism	supported	Six-cycle contract and one-million-cycle integrated CXXRTL run with zero visible mismatch; 19/19 gates	V4 pre-board result, not V5 or board measurement
Target-device feasibility	supported	Integrated V4 minimum Fmax 39.137 MHz; source-bound static core 41.024–42.212 MHz; bounded resources	Open-tool estimate; 222.222 ns is a contract-clock estimate
Teacher-student finite-model extension	supported	Effective-fidelity lifetime was $2.36 \times / 1.58 \times$ fixed parameters; 95-scalar student retained most teacher gain with bit-exact RTL	Ten-cycle component/model evidence; not LER, wall-clock lifetime, or official GQF superiority

Metric / property	Status	Evidence-supported statement	Boundary
Historical CNN residual LER	conditional	Hybrid Residual-B reduced T24 final LER by 1.75–2.89% relative to UKF in four frozen scenarios	Old mock-backed software-HIL; 0/16 current learned families are same-task eligible
Official GQF superiority	not established	Exact-source checklist 0/15 and all 13 matched metrics null	No reproduction or comparison with Puviani <i>et al.</i>
Measured board / external speed	not established	Board latency, jitter and power are null; 18 external rows contain 0 exact same-task comparator	No measured latency, raw-ns ranking, fastest, or SOTA claim

8 Negative results that define the design

The contract-centric narrative is credible only if its negative evidence is kept visible.

1. **The CNN-centric strong branch failed.** The current main system cannot rely on a learned winner, and the historical CNN does not meet the new matched execution contract.
2. **Static fallback and freeze-all adaptation failed.** All 38 candidate tuples were NO-GO for both V2 and V3 because a frozen trusted state became stale under persistent change. Continuous validated shadows were an evidence-driven correction, not cosmetic refactoring.
3. **Fallback is not free.** The earlier uncertainty gate helped the aggregate OOD mixture but harmed compound OOD and slightly harmed nominal operation. In the later formal matrix, registered non-inferiority passed, but tail fallback/unnecessary signals reached 59–96% and 59–95%.
4. **The smooth primary gate is not a leaderboard.** ROUTE-A passed the locked EWMA contrast but was worse than static and Window MAP, with negative oracle-gap closure and a gain concentrated in periodic drift.
5. **V5 lacked deployable causal headroom.** Most diagnostic oracle gain came from truth-privileged estimation and activation. The expanded action family added only 0.02549%, so implementation was correctly stopped.
6. **Positive auxiliary results do not repair a failed system gate.** Multimode CPD and the two-GKP-CNOT reproduction are useful independent results; using them to relabel V5 as successful would be cross-task leakage.
7. **Hardware evidence remains layered.** CXXRTL, synthesis, P&R, and a six-cycle contract do not imply a bitstream, transport integration, board timing, power, or quantum-device experiment.

9 Terminal verdicts and remaining gates

The main simulation/pre-board promotion decision is already terminal: V4 is restricted and V5 was stopped. Remaining work can improve integrity, reproducibility, or measured-hardware coverage, but it cannot silently change those verdicts.

Table 11: Current terminal states and remaining evidence gates.

Gate	Current requirement	Consequence for writing
V4 final gate (T6.9.3)	Terminal NO-GO; restricted simulator/pre-board draft only	V4 may support a restricted simulator/pre-board systems account, not broad decoder superiority
V5 final gate (T6.15.5)	Terminal NO_GO_V5_EARLY_HEADROOM_STOP	No V5 performance or hardware result may be synthesized from V4 artifacts
Auxiliary integrity (T6.19.3)	Terminal PASS_AUX_COMPARISON_INTEGRITY: 206 cells, 24/24 gates and mutations, 162 joint regressions	Allows within-lane secondary reporting only; V5 NO-GO and board Blocked remain unchanged
Real board (T6.9.2)	Bitstream, transport, correctness, latency distribution, deadline miss, resources and power on the named board	Hardware language remains CXXRTL/synthesis/P&R estimate only
Phase 7 manuscript gate	Every abstract/conclusion sentence maps to a task-local artifact and evidence layer	The formal paper should be an evidence-gated/negative-results account or a narrower independent algorithm paper

10 Discussion

The main contribution at this cutoff is not a winning adaptive decoder. It is an evidence-gated research object in which admission, causal observability, tail intervention, action capacity, fixed-point execution, and hardware maturity can fail independently. This organization reuses the affine parameter surface, frozen CNN result, teacher/student work, fixed-point pipeline, and FPGA-oriented bank machinery without allowing any one extension to carry unsupported system-wide claims.

The smooth formal result illustrates both the value and the cost of this framing. Because EWMA was selected before formal access, the positive primary interval is meaningful for that contrast. Because static and Window were also reported, the same experiment prevents a misleading best-method claim. The periodic family reveals a narrow routing opportunity, while the other families show that extra policy complexity can be neutral or harmful. The V5 audit strengthens this interpretation: large truth-oracle gains did not survive causal observation, and adding actions did not address the dominant regret. Regime awareness is useful only when the observations contain enough deployable information to select a better action.

The hardware line is similarly bounded. Fixed cycles, packed words, atomic banks, mutation-tested failures, P&R estimates, and million-cycle bit parity are stronger than a software-only latency sketch. They are nevertheless one layer below a measured control plane. The eventual paper should report core, transport, source-to-action, and end-to-end latency separately, and should compare only with external work sharing a compatible task and timing definition. The normalized external table currently has no exact same-task comparator, which is a reason to leave the raw-nanosecond ranking empty rather than to force an incommensurate speed claim.

The tail matrix illustrates an important distinction between bounded safety and performance advantage. V4 passed every registered degradation margin against locked EWMA, including nominal controls, so the policy did not reintroduce the earlier compound failure. Yet most contrasts are exact ties, static MAP is far better under calibration shift, and safety signals are active through much of the tail interval. The honest interpretation is auditable locked-baseline non-inferiority with high intervention cost. The subsequent integer qualification preserved the execution fields over one million cycles, but did not change the algorithmic ordering or populate any board-measured metric.

Phase 6C shows why evidence lanes should remain separate rather than be hidden. The multimode posterior-weighted CPD result has a large, consistent LER and tail gain; the CNOT reproduction validates an analog-ML effect; AQEC replay is negative; learned and GQF same-task comparisons are unavailable; and FPGA comparison is task-incommensurate. Together they form a useful secondary map, not a scalar score and not a substitute for the failed V5 gate.

11 Reproducibility and evidence availability

Every numerical statement in the main figures is regenerated from repository JSON or source-data CSV rather than transcribed from presentation text. The Python entry point is `docs/figures/make_route_a_manuscript_figures.py`. It exports editable SVG, vector PDF, print-resolution LZW-TIFF, review PNG, and a 58-row plotted-data table under `docs/figures/route_a_manuscript_20260720/`. The accompanying manifest records SHA-256, byte size, task-local source path, figure contract, and every output hash. The two reused Chinese presentation figures retain their original source-data and generator paths and are labelled as archival interface/model evidence rather than current promotion results. The repository-generated T6.19.3 atlas separately binds all 206 auxiliary cells and its SVG/PDF/TIFF/PNG bundle to the terminal integrity artifact.

Randomized comparisons report the independent cluster unit, paired interval, and decision denominator in their captions or tables. Hardware-facing tables retain the evidence layer—fixed-point, CXXRTL, synthesis, P&R estimate, or board measurement—and preserve unmeasured board fields as null. The appendix maps each result to its human-readable report and machine artifact.

12 Conclusion

The current evidence supports a contract-centric, regime-aware dual loop as a disciplined way to separate GKP decoding performance, tail-safety control, learning extensions, and deterministic execution. It does not support the broad adaptive-decoder paper originally sought. V4 has a narrow positive contrast against locked EWMA and a periodic-family discovery, but static and Window MAP are better in aggregate and abrupt/OOD results provide non-inferiority rather than tail advantage. V5 was correctly stopped because deployable and incremental action-space headroom were too small.

The remaining positive claims are explicit and task-local: observed-only posterior weighting improves the independent multimode CPD task; analog ML improves the reproduced two-GKP-CNOT task; the compressed student retains most teacher gain in a finite model; and V4/static-MAP cores satisfy deterministic pre-board execution contracts. None establishes measured FPGA speed, official GQF superiority, or a physical quantum-memory improvement. This transparent partition—including the negative and null cells—is the result that can support a formal evidence-gated systems paper or a narrower algorithm paper.

A Primary repository evidence map

Topic	Human-readable source	Machine/source-data source
Claim and role contract	<code>docs/route_a_claim_contract.md</code>	<code>docs/t6_5_1_route_a_claim_contract.json</code> and CSV
Unified execution contract	<code>docs/unified_execution_contract.md</code>	<code>docs/t6_5_2_unified_execution_contract.json</code> and CSV
Formal preregistration	<code>docs/route_a_preregistration.md</code>	<code>docs/t6_5_3_route_a_preregistration.json</code> and CSV
Comparator qualification	<code>docs/unified_comparator_runner.md</code>	<code>docs/t6_6_1_unified_comparator_runner.json</code> and CSV
V4 policy	<code>docs/regime_aware_safe_policy.md</code>	<code>docs/t6_6_2_regime_aware_safe_policy.json</code> and CSV
Posterior/threshold lock	<code>docs/route_a_posterior_calibration.md</code>	<code>docs/t6_6_3_route_a_posterior_threshold_lock.json</code> and CSV
Smooth formal matrix	<code>docs/route_a_smooth_formal.md</code>	<code>docs/t6_7_1_smooth_formal_matrix.json</code> and 498,240-row CSV

Topic	Human-readable source	Machine/source-data source
Abrupt/OOD formal matrix	docs/route_a_tail_formal.md	docs/t6_7_2_abrupt_ood_tail_formal_matrix.json and 686,104-row CSV
V4 integrated qualification	docs/route_a_integrated_rtl_qualification.md	docs/t6_7_3_route_a_integrated_rtl_qualification.json and raw traces
V4 final evidence gate	docs/route_a_final_evidence_gate.md	docs/t6_9_3_route_a_final_evidence_gate.json and source-data CSV
V5 causal headroom / stop	docs/route_a_causal_headroom.md; docs/route_a_v5_final_evidence_gate.md	docs/t6_10_1_causal_headroom.json; docs/t6_15_5_route_a_v5_final_evidence_gate.json
Two-GKP CNOT reproduction	docs/noh_cnot_ci_ml_reproduction.md	docs/t6_17_2_noh_cnot_ci_ml_reproduction_source_data.csv
Multimode CPD	docs/multimode_posterior_weighted_cpd.md	docs/t6_18_3_multimode_posterior_weighted_cpd.json and CSV
Current pre-board profiles	docs/project_preboard_profiles.md	docs/t6_19_1_project_preboard_profiles.json and raw P&R reports
External FPGA refresh	docs/external_fpga_decoder_refresh.md	docs/t6_19_2_external_fpga_normalization.json and CSV
Six-lane auxiliary integrity	docs/secondary_comparison_atlas.md	docs/t6_19_3_secondary_evidence_integrity.json and one-row-per-cell CSV
Legacy frozen result	docs/paper_materials/paper_simulation_result_table_pack.md	runs/p4_benchmark/T24_formal_software_revalidation_20260510_200743/comparison.csv
Target P&R	docs/target_device_synthesis.md	docs/t5_5_2_target_device_synthesis.json and raw reports
Student Pareto	docs/precision_resource_performance_pareto.md	docs/t5_5_3_precision_resource_pareto.json and equivalence report
Long RTL qualification	docs/long_rtl_qualification.md	docs/t6_2_2_long_rtl_qualification.json and CSV
Dual-loop presentation schematic	docs/figures/ppt_summary_20260716/drift_adaptive_architecture_cn.pdf	docs/figures/make_ppt_summary_figures.py
Teacher-student presentation figure	docs/figures/ppt_summary_20260716/effective_fidelity_lifetime_cn.pdf	docs/t4_4_4_teacher_student_gain_retention_source_data.csv and figure generator
Evidence-bounded manuscript figures	docs/figures/route_a_manuscript_20260720/	Source-data CSV and SHA-256 manifest in the same directory
Auxiliary atlas figure	docs/figures/t6_19_3_secondary_comparison_atlas.pdf	T6.19.3 source-data CSV and integrity JSON

B Claim wording guardrails

- Allowed now: “improves over the pilot-locked EWMA baseline on the untouched smooth aggregate, with the family-level discovery concentrated in periodic drift.”
- Allowed now: “passed the pre-registered abrupt/OOD and nominal non-inferiority gates relative to locked EWMA; most core contrasts were ties.”
- Not allowed now: “outperforms static GKP decoding”, “best adaptive decoder”, “improves tail LER”, “outperforms under OOD”, or “CNN provides the main gain.”
- Required now: state that V5 was stopped before implementation because deployable and incremental action-space headroom failed the entry gates.
- Allowed now: “observed-only posterior-weighted CPD improved aggregate, worst-window, and CVaR metrics in the independent multimode matched task.”
- Not allowed now: use the multimode CPD result to claim V5 or full-system promotion, or collapse Phase 6C lanes into a global ranking.
- Allowed now: “the fixed-point core has a six-cycle contract and passed board-independent CXXRTL and target-device P&R qualification.”
- Not allowed now: “measured sub-microsecond FPGA decoder”, “faster than existing FPGA QEC decoders”, or “hardware validated.”
- Allowed now: “the distilled student retained most teacher gain in the project finite model and fit a selected pre-board design point.”
- Not allowed now: “reproduces or exceeds Puviani NMF” or “proves a universal non-Markovian memory advantage.”
- Required now: board latency, jitter and power remain `null`; the six-cycle 222.222 ns value at 27 MHz is a contract-clock estimate.

References

- [1] D. Gottesman, A. Kitaev, and J. Preskill, “Encoding a qubit in an oscillator,” *Physical Review A* **64**, 012310 (2001).
- [2] A. L. Grimsmo and S. Puri, “Quantum error correction with the Gottesman–Kitaev–Preskill code,” *PRX Quantum* **2**, 020101 (2021).
- [3] E. H. Chen *et al.*, “Calibrated decoders for experimental quantum error correction,” *Physical Review Letters* (2022).
- [4] V. Sivak, M. Newman, and P. Klimov, “Optimization of decoder priors for accurate quantum error correction,” *Physical Review Letters* **133**, 150603 (2024).
- [5] M. Puviani, S. Borah, R. Zen, J. Olle, and F. Marquardt, “Non-Markovian feedback for optimized quantum error correction,” *Physical Review Letters* **134**, 020601 (2025).
- [6] V. Sivak *et al.*, “Real-time quantum error correction beyond break-even,” *Nature* (2023).